

CivicLegal AI

AI-Powered Citizen Legal Rights & Government Services Navigator

Project Proposal | April 2026 | github.com/dp2426-NAU/Civiclegal

Executive Summary

CivicLegal AI is a Retrieval-Augmented Generation (RAG) web application that helps citizens navigate their legal rights and discover government benefit programs using plain-language queries. The system delivers grounded, cited legal information (not advice) through a 5-layer anti-hallucination pipeline, making legal knowledge accessible to underserved communities without requiring any legal expertise from the user.

Problem Statement

The United States faces a significant 'legal gap': 57 million Americans experience civil legal problems annually, yet only ~20% receive adequate assistance. Legal information is fragmented across court databases, federal registers, and government portals, and the language used is inaccessible to most citizens. Simultaneously, millions of eligible citizens miss out on government benefit programs simply because they are unaware of them.

Proposed Solution

CivicLegal AI provides three core capabilities:

1. Legal Rights Navigator — plain-language question answering with citations
2. Benefits Finder — automated eligibility matching for 10 federal programs (SNAP, Medicaid, CHIP, WIC, TANF, Section 8, SSI, SSDI, EITC, LIHEAP)
3. Document Upload — contextual analysis of uploaded legal documents (PDF/DOCX)

All responses include citations, confidence scores, and mandatory legal disclaimers. A full demo mode operates without any API keys.

Technical Architecture

Frontend: Streamlit (4 tabs with sidebar disclaimer)

Backend: FastAPI with endpoints /health, /legal-query, /benefits-match, /upload-document

RAG Pipeline: LangChain + ChromaDB (vector store) + FAISS + BM25 hybrid retrieval

Reranking: ms-marco-MiniLM-L-6-v2 cross-encoder

LLM: GPT-4 (primary) → Mistral 7B via Ollama (fallback) → Demo mode (no key needed)

Embeddings: sentence-transformers/msmarco-MiniLM-L-6-v2

Document Processing: PyMuPDF (PDF), python-docx (DOCX)

Chunk Size: 500 tokens, Overlap: 75 tokens, Temperature: 0

Anti-Hallucination System

CivicLegal AI uses a 5-layer hallucination prevention guard:

Layer 1 — Refusal detection: identifies uncertainty signals and penalises confidence

Layer 2 — Grounding check: cosine similarity ≥ 0.50 against retrieved context required

Layer 3 — Overconfidence detection: flags absolute claims, caps confidence at 70%

Layer 4 — Citation validation: filters source references not grounded in the answer

Layer 5 — Disclaimer injection: mandatory disclaimer appended to every response

Temperature is set to 0 throughout the pipeline to ensure deterministic, factual output.

Data Sources & APIs

CourtListener API — Federal and state court opinions

Congress.gov API — Federal legislation (118th Congress)

Regulations.gov API — Federal regulatory documents

Benefits.gov Dataset — Government benefit program data

CUAD Dataset — Contract analysis and understanding

Sample Legal Documents — Included demo documents (tenant rights, constitutional rights, ADA)

All API clients implement graceful degradation: if a key is missing or a service is down, clearly labelled demo data is returned automatically.

Benefits Matching Engine

The benefits matcher uses 2024 Federal Poverty Level (FPL) guidelines to determine eligibility across 10 federal programs. Inputs include household size, annual income, state, presence of dependent children, disability status, veteran status, and age.

Programs covered: SNAP, Medicaid, CHIP, WIC, TANF, Section 8 Housing Voucher, SSI, SSDI, Earned Income Tax Credit (EITC), LIHEAP.

Each result includes: eligibility status, reason for determination, required documents, agency name, and a direct link to apply.

Deployment

Local Development:

Backend: `uvicorn backend.main:app --port 8000`

Frontend: `streamlit run frontend/app.py --server.port 8501`

Cloud Deployment (Render.com):

The included `render.yaml` deploys both services automatically.

Connect the GitHub repository at github.com/dp2426-NAU/Civiclegal to Render, and both the FastAPI backend and Streamlit frontend deploy with one click.

Environment Variables:	OPENAI_API_KEY,	CONGRESS_API_KEY,
REGULATIONS_API_KEY,		
COURTLISTENER_API_KEY,	BACKEND_URL,	USE_DEMO_DATA,
OLLAMA_BASE_URL		

Testing

The project includes a pytest test suite covering:

- Health endpoint — status, fields, disclaimer presence
- Legal query — demo response, short-question rejection, disclaimer
- Benefits matcher — FPL calculation, program eligibility logic, required fields
- Hallucination guard — all 5 layers: refusal, similarity, overconfidence, citations, disclaimer

Run tests: `pytest tests/ -v`

Legal & Ethical Compliance

CivicLegal AI is designed with the following safeguards:

- The system explicitly provides information, never legal advice
- No attorney-client relationship is created
- All AI-generated content is labelled with confidence scores
- Demo data is clearly marked [DEMO DATA – NOT LEGAL ADVICE]
- Users are directed to licensed attorneys and legal aid organisations
- Free legal resources are linked throughout the application

LEGAL DISCLAIMER: CivicLegal AI provides general legal INFORMATION only — NOT legal advice. Consult a licensed attorney for advice specific to your situation.

Timeline

Week 1 — Architecture design, system diagram, API specification

Week 2 — FastAPI backend, benefits matcher, API clients

Week 3 — RAG pipeline: ChromaDB, hybrid retrieval, cross-encoder reranker

Week 4 — Streamlit frontend with all 4 tabs

Week 5 — 5-layer anti-hallucination guard system

Week 6 — pytest test suite, full documentation

Week 7 — Render.com deployment, GitHub push, final review

GitHub Repository

Repository: <https://github.com/dp2426-NAU/Civiclegal>

Structure: `antigravity/civiclegal-ai/`

License: MIT

Legal Disclaimer: LEGAL DISCLAIMER: CivicLegal AI provides general legal INFORMATION only — NOT legal advice. Consult a licensed attorney for advice specific to your situation.